Same Bytes, Different Scores: What a Code-Agent Leaderboard Measures

Evidence from the 2026 Gemma 4 Developer Agent competition

Wang Chen and **Youtao Liu** · South China University of Technology

Abstract

Leaderboards for coding-agent competitions are read as capability rankings, and low scores as capability failures. Merging two independently collected observation sets from the 2026 competition, we argue one position supports neither reading; we analyse one archive and cite the other. Three archives with identical SHA-256 (6ff73d10…, 3,713 B) were scored 0.22, 0.20 and 0.15; two submissions advertising 0.22 and 0.24 measured 0.15, their claims already inconsistent; across 4,521 teams score co-varies with submission count (Spearman 0.5841), median counts 4→17.5, and local evaluations resolving 55.6–94.4% do not predict a public 0.15. The partner archive, cited as background, reports pathological tool-call failures that are not diagnostic. A score is one draw from a partly competitor-chosen sampling process; reports should state attempts, distributions, difficulty strata and cost.

Keywords: coding agents; leaderboard validity; evaluation reliability; reproducibility

TL;DR

- Three archives with identical SHA-256 scored 0.22, 0.20 and 0.15 on the public leaderboard.
- Two public claims of 0.22 and 0.24 measured 0.15; the same bytes resubmitted stayed unscored after 11.8 h.
- Across 4,521 teams score co-varies with submission count (ρ = 0.58; medians 4→17.5): one score is one draw, not a capability estimate.

1 Introduction

Competitions publish a leaderboard number beside a downloadable package, and the community reads it as a capability statement: a higher score means a better agent, a failed run an agent that could not do the task. Our data suggest it fails on both sides.

On the capability side, the number is a draw. Three teams published archives with identical SHA-256 (6ff73d10…, 3,713 B) and were scored 0.22, 0.20 and 0.15 (LIN-03…LIN-05); two submissions advertising 0.22 and 0.24 measured 0.15, their claims already contradicted by other public statements (SUB-05, PROV-01…PROV-03); identical bytes resubmitted stayed unscored after 11.79 h (SUB-03b), so a leaderboard mixes evaluated and unevaluated attempts.

On the failure side, an observed failure is not a diagnosis: the partner team's 129-trace archive reports pathological rejection loops while affected tasks resolve at rates close to the rest (A-01…A-03, A-15); we cite it as background, not as our result.

Our four measured contributions. (1) *Identical bytes, different scores* (LIN-03…LIN-05). (2) *Inconsistent advertisements and a pending measurement* (SUB-01…SUB-05, PROV-01…PROV-03). (3) *A best-of-n sampler:* 4,521 teams, ρ = 0.5841, medians 4→17.5, 43% of teams ≤3 submissions (LB-00…LB-02, LB-band-15-dist, LB-band-24-dist). (4) *Local evaluation does not predict:* two 18-task blocks, pass rates 55.6–94.4%, public

0.15 (LOC-d1-ALLARMS, LOC-v10-ALLARMS, X-LOCAL). Claims are descriptive; we identify no causal effect and assume no transfer beyond this competition, model and split.

2 Related Work

Repository-level benchmarks report single-run pass rates (Jimenez et al., 2024; Yang et al., 2024; Xia et al., 2024); single runs are insufficient (Miller, 2024; Agarwal et al., 2021; Bouthillier et al., 2021). Self-refinement reports gains, but models cannot self-correct reliably without external feedback (Madaan et al., 2023; Shinn et al., 2023; Huang et al., 2024; Stechly et al., 2024; Kamoi et al., 2024). Function-calling benchmarks separate wrong-function from wrong-argument errors (BFCL; Qin et al., 2023); leaderboards are sensitive to sampling and reporting choices (Chiang et al., 2024; Singh et al., 2025).

3 Setting and Measurement

3.1 The competition

The 2026 Gemma 4 Developer Agent competition asks for a *declarative agent package* (configs and prompts compiled by the `swegemma` harness), served by one permitted Gemma 4 model on four L4 GPUs with 32,768-token context (SET-HARNESS-HW, SET-HARNESS-CTX, SET-HARNESS-SINGLE), and unpacking below 3 GiB (SET-HARNESS-SIZE). A task is **resolved** iff the test process exits 0 with at least one passed test and no failures or errors. Development has 129 tasks (SET-DEV), evaluation 58 (SET-PUBLIC); one task is worth 1/58 ≈ 0.0172 (SET-STEP), scores truncated to two decimals. A submission evaluates all 58 tasks inside a 12-hour global budget (SET-HARNESS-BUDGET) with per-task limits (SET-BUDGET: 8 min, 28 calls, 80 turns). One submission per day with an exclusive PENDING window (SET-HARNESS-SUBLIMIT); the reported score is the best public score (Figure 4).

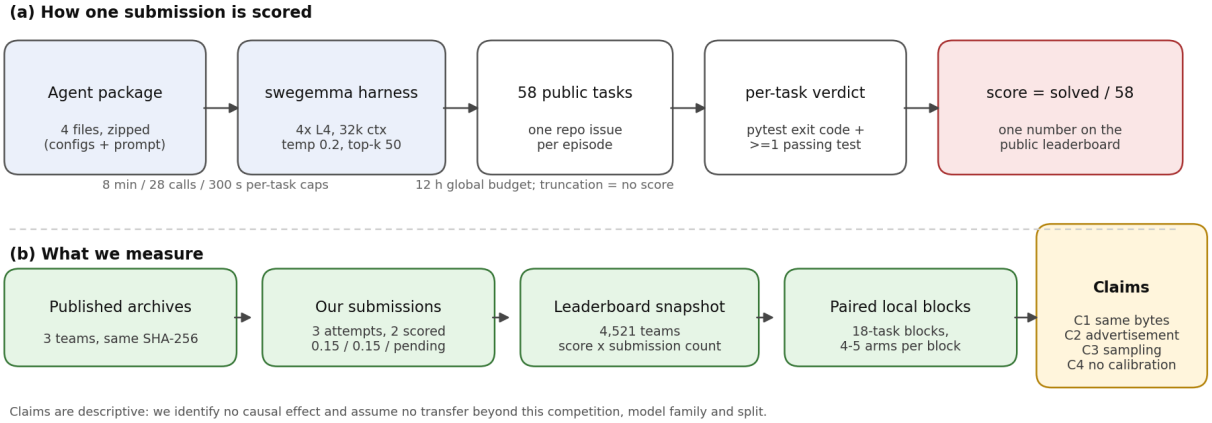


Figure 4. How one submission is scored (a), and the four measurement sources behind our claims (b).

3.2 Data sources and authorization

Archive A (partner team). First-run traces on the 129-task development set (AutoDL), with a later L4 rerun of one bundle: tool calls, rejections, budget counters, outcomes. Retrospective and filtered; the partner team authorized its use and reported the numbers we cite; we did not recompute them. **Archive B (our data).** Nine public packages clustered into seven payloads from file and zip SHA-256 (LIN-01, LIN-02); a third-party 2026-11-03 snapshot of 4,521 leaderboard teams (LB-00); two 18-task

blocks with six and five variants (SET-EVAL-TASKS, LOC-d1-NARMS, LOC-v10-NARMS); and our submission records (SUB-01…SUB-05).

3.3 Identities and events

A **rejection** is a schema-level error where a required argument was absent; an **adjacent exact retry** repeats the same tool and byte-identical arguments; a **streak** is a maximal run of consecutive rejections; a **charged call** counts against the budget. We compute the **zip digest**, the **payload signature** over four binding files, and a **lineage**; identity is necessary but not sufficient (LIN-12, LIN-15).

3.4 Budgets and cost

Two budgets apply: per-task and submission. Rejected calls are not charged but consume turns and wall-clock, so a failing run can look cheap yet exhaust time (18 + 5 time-cap, 12 + 15 tool-cap hits; LOC-d1-CAPS, LOC-v10-CAPS). 58 tasks at the local 8-minute budget would consume 464 minutes before setup and verification — an upper bound under a uniform per-task budget; the six batches used 902.74 wall-clock minutes, 30.10 GPU-quota hours on 4×L4 (SET-WALL; SET-WALL-D1, SET-WALL-V10).

4 Results

4.1 Tool-call failures are not diagnostic (partner-reported)

The partner team reports 70 missing-argument rejections among 340 recorded editor requests (20.6%; A-01, co-author manuscript, lines 41/53), concentrated in 14 of 129 tasks (A-02). Among the 14 affected tasks 6 resolved (42.9%) and among the other 115, 45 (39.1%; A-03): close rates, no established difference. Its longest streak was 12 and it resolved 51/129 (A-04, A-05). Case `fastapi_13713`: 54 rejections in two streaks of 18 and 36, 52 adjacent exact retries, then its 80-turn limit with 26 charged calls, unresolved (A-06, A-07, A-08). Case `fastapi_14986`: 41 consecutive rejections, 40 exact retries, ~289 s and 13 charged calls against an 8-minute limit, unresolved (A-09, A-10, A-11). A recovery instruction removed the loop without changing outcomes: on eight shared tasks rejections fell from 60 to 1, both variants resolved 0 of 8, though it used more calls and time (182→217; 37.8→46.0 min; A-12); a third variant, on a partly overlapping eight-task set, resolved 1/8 with 21/38 rejections (A-13). The same v5 bundle produced 5 missing-argument rejections across 24 recorded editor requests on the AutoDL host (20.8%) versus 42 in 63 on Kaggle L4 (66.7%), with zero resolutions in both on the eight-task core subset; the bundle hashes match, so this is variation across recorded runs, not a hardware estimate (A-14). Follow-up cohorts were drawn from previously failed tasks and differ in size (A-15), so we do not claim it is useless.

4.2 Identical bytes, different scores

Three archives with identical SHA-256 (6ff73d10…, 3,713 B) were scored 0.22, 0.20 and 0.15 (LIN-03, LIN-04, LIN-05). They share payload lineage 397f971bfc0f; the spread is 0.07, about four of 58.

Our re-submissions show inconsistent advertisements and one incomplete evaluation. Ref `58056163`'s public notebook gave three contradictory self-descriptions — title `LB 0.24`, cell note `v6 (0.22 config)`, author card `0.18` (PROV-01) — and measured 0.15; no bundle digest was logged (PROV-01b), so we do not call it byte-identical to any scored payload. Ref `58077152`'s description read "byte-identical to … davis0178/gemma-4-agent-baseline v19 …, public LB 0.22" (PROV-02), but the bytes actually submitted are the v24 bundle (sha 481c0468…, 2,404 B); the davis-v19 payload (e387ba9b…, 2,767 B) was never submitted. It measured 0.15. Ref `58111401` uploaded the same v24 bytes and, after

11.79 h, had no score (SUB-03b, PROV-03). Delays are part of the measurement conditions (thread #747519); a late or absent score is a missing measurement, not a low one.

Table 1. Lineages and public scores.

Evidence	Artifact	Public score
L1 (LIN-03, LIN-03b, LIN-04, LIN-05)	zip sha 6ff73d10…, 3,713 B; 4/4 binding files identical, three teams	0.22 / 0.20 / 0.15
L2 (LIN-06), L4 (LIN-08), L5 (LIN-09), L6 (LIN-10)	not submitted by us	0.20; 0.20; 0.12 (best 0.18); 0.13 (best 0.20)
L3 (LIN-07, LIN-12)	our first pack: 4/4 members equal, zip differs	0.17 (best 0.20)
L4 (LIN-13), L7 (LIN-11, LIN-14)	our packs match on 3/4 members; L7 has no score	0.20; none

Fig. 1 Same bytes, different scores: public payload lineages and what they scored

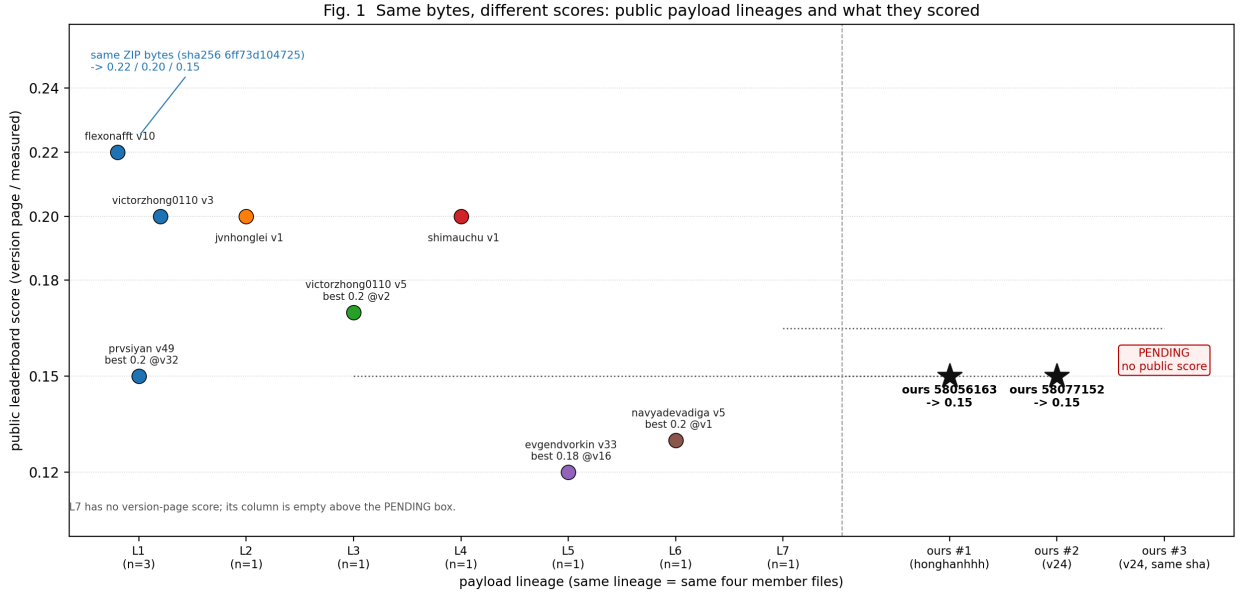


Figure 1. Lineages and our submissions by public score.

Table 2. Submission provenance (from `submission-log.csv`).

Ref	Public score	Status
58056163	0.15	COMPLETE
58077152	0.15	COMPLETE
58111401	—	PENDING

4.3 The leaderboard behaves like a best-of-n sampler

Across 4,521 teams (LB-00), score and submission count are associated (Spearman 0.5841; LB-01). Median submission counts rise from 4 (0.15 band, n = 648) to 7 (0.17, n = 812), 10 (0.18, n = 703), 13.5 (0.20, n = 288), 17 (0.22, n = 63) and 17.5 (0.24, n = 8); by 0.22 no team has ≤3 submissions (LB-band-15-dist, LB-band-17-dist, LB-band-18-dist, LB-band-20-dist, LB-band-22-dist, LB-band-24-dist). At the other end, 43% of teams made ≤3 submissions (1,943 of 4,521; LB-02). 16 distinct scores appear, all k/58 truncations; 0.25 and 0.27 are absent (LB-03, LB-04).

This is consistent with repeated sampling. With a single-draw success probability p = 9/58 ≈ 0.155, the rate implied by our 0.15 (X-IMPLIED), 13/58 (0.22) has per-draw probability ≈ 0.106 (BIN-01), about 9.5 draws (BIN-04); 14/58 gives ≈ 0.057 (17.5 draws; BIN-02, BIN-05), and 17/58 ≈ 0.006 (174

draws; BIN-03, BIN-06). The pattern does not establish a single p: the 0.29 team reached 17/58 with 18 submissions (LB-band-29-dist), far fewer than the model needs. Submission count is a first-order covariate: ignoring it compares sampling budgets as much as agents.

Table 3. Leaderboard bands (2026-11-03 snapshot).

Public score	Teams	Median submissions	Teams ≤3
0.15	648	4	270
0.17	812	7	217
0.18	703	10	117
0.20	288	13.5	19
0.22	63	17	0
0.24	8	17.5	1
0.29	1	18	0

Fig. 2 The public LB as a best-of-n sampler: score vs number of submissions (2026-11-03 snapshot, n=4521)

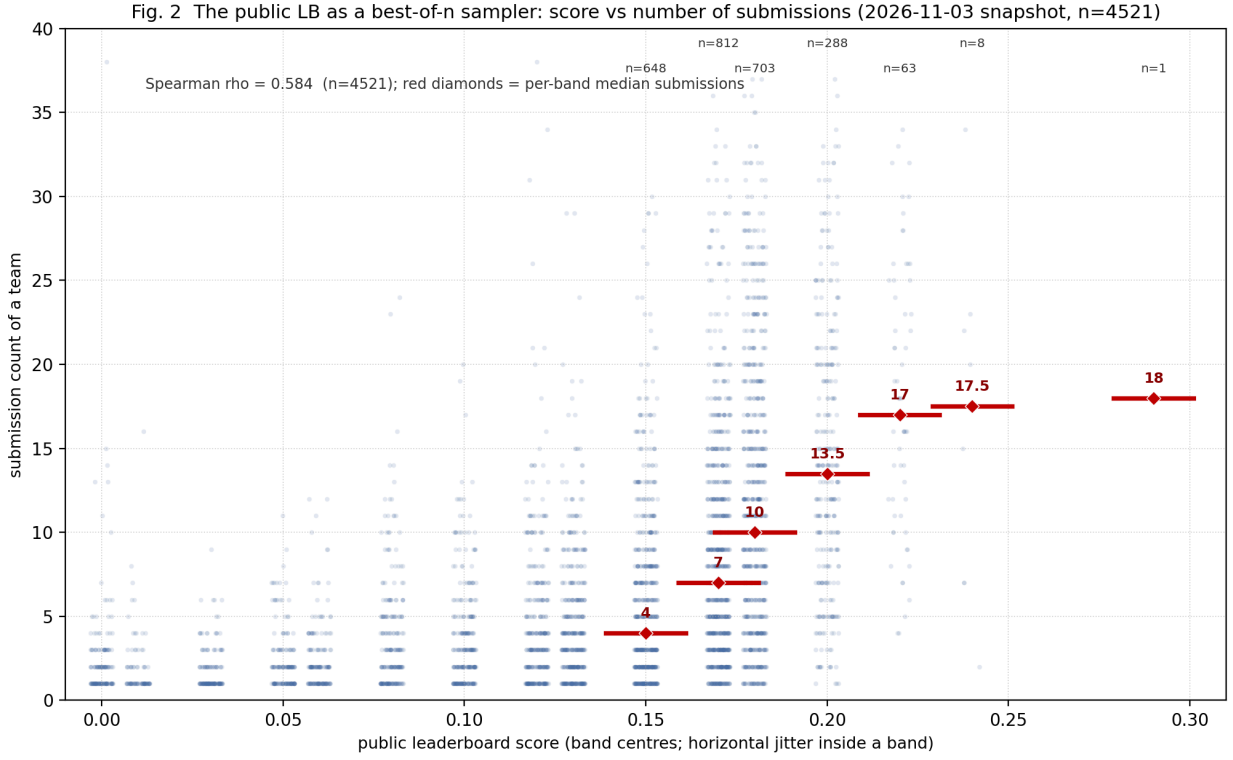


Figure 2. Score band vs submission count, band medians and waiting-time range.

4.4 A paired local evaluation does not predict the leaderboard

Two paired blocks ran on the official harness: in D1 six arms on the same 18 development tasks with identical budgets, differing in sampling and payload; in V10 five arms on another 18 tasks (SET-EVAL-TASKS).

The blocks mostly measure *task* difficulty, not arm quality. In D1, 10 of 18 tasks were resolved by every arm and 7 by none; in V10, 14 of 18 by every arm and none by none (LOC-d1-ALLARMS, LOC-v10-ALLARMS). Between-arm differences are small: disagreement is at most one task in D1 (LOC-d1-DISAGREE) and four in V10 (LOC-v10-DISAGREE). Per-arm counts are out of 18 tasks; 15 passed gold in D1 (LOC-d1-GOLD), all 18 in V10 (LOC-v10-GOLD). Empty patches were rare (3/108 D1, 1/90 V10; LOC-d1-EMPTY, LOC-v10-EMPTY).

The local level is far above the public level. The local arms are built from the v24 and davis-v19 payload families; the v24 submission measured 0.15 publicly (SUB-02), while the arms resolve 55.6–61.1% of development tasks in D1 and 83.3–94.4% in V10 (X-LOCAL), with pooled pass rates 59.3% and 88.9% (LOC-d1-POOLED, LOC-v10-POOLED). A team tuning here would see most development tasks solved; the leaderboard places it at 0.15 (9 of 58), 3.6–6.1× lower (X-IMPLIED). Since the hidden distribution is unknown, we treat the gap as a measurement fact: the evaluation helps debugging but does not predict the public band, and a one- or two-task difference is not a ranking.

Table 4. Local paired evaluation. Ids: `LOC-D1-<A..F>-<variant>`, `LOC-V10-<A,C,D,E,G>-<variant>`.

Block	Arm	Difference	Resolved
D1	A (LOC-D1-A)	v24 byte-exact	11/18
D1	B (LOC-D1-B)	v24 safe-55	10/18
D1	C (LOC-D1-C)	davis v19 byte-exact	11/18
D1	D (LOC-D1-D)	v19 safe-55	11/18
D1	E (LOC-D1-E)	v24 temp 0.2	11/18
D1	F (LOC-D1-F)	davis temp 1.0	10/18
V10	A (LOC-V10-A)	v24 byte-exact	16/18
V10	C (LOC-V10-C)	davis v19 byte-exact	15/18
V10	D (LOC-V10-D)	v19 safe-55	17/18
V10	E (LOC-V10-E)	v24 temp 0.2	16/18
V10	G (LOC-V10-G)	eager submission	16/18

Fig. 3 Local paired evaluation: per-arm PASS and mean wall time (18-task blocks)

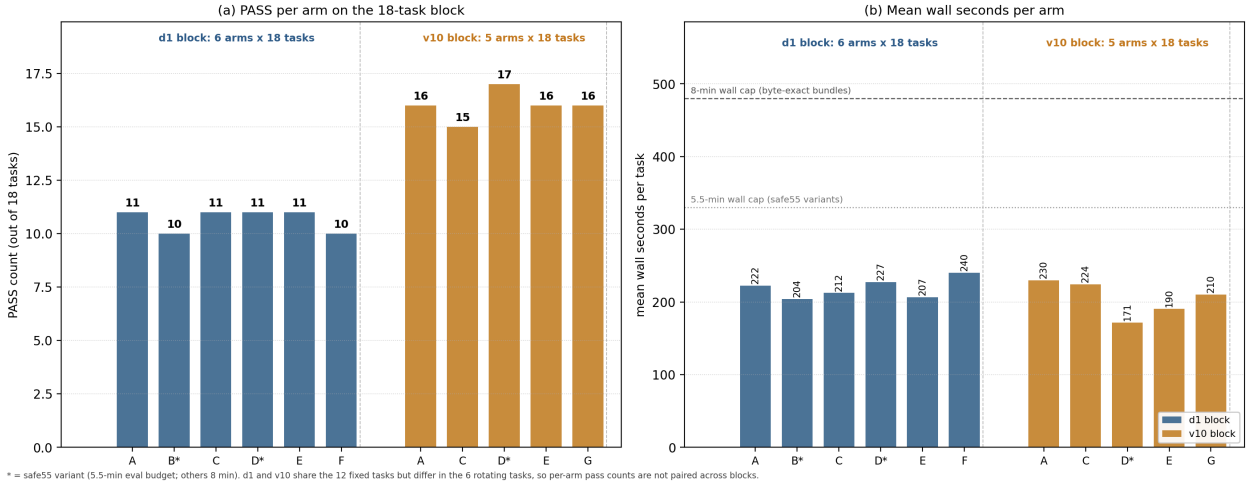


Figure 3. Per-task resolution matrices for the D1 and V10 blocks.

4.5 What did not work

- Local 18-task blocks cannot separate four or five arms: all arms pass 10/18 in D1 and 14/18 in V10 (LOC-d1-ALLARMS, LOC-v10-ALLARMS).
- 55.6–94.4% local pass rates do not predict the public 0.15 (X-LOCAL).
- Prompt and payload interventions left outcomes unchanged in the partner archive (A-12, A-13, A-14).

5 Discussion and Limitations

What a score measures. The public score depends on artifact, sampling policy, submission count and task set; the leaderboard reports one. Whether the pending evaluation would have scored the same bytes differently is unknown.

A reporting protocol. (1) Report each score with attempts, median and IQR (Agarwal et al., 2021; Bouthillier et al., 2021). (2) Stratify by difficulty: an all-pass/all-fail core makes an unstratified sample wasteful. (3) Report cost and sampling: GPU-hours, wall-clock, rejections, submission history. (4) Score a pre-declared final submission, or the mean of a fixed number of independent runs, rather than the maximum over unlimited attempts, and publish the attempt budget beside the score.

Failure analysis. A high-frequency failure can be removed at the prompt level without changing resolution; a leaderboard-level failure cannot be diagnosed from status alone. The loop's causal role is unestablished: both variants resolved 0 of eight shared tasks, eight are too few, and the cohorts came from historical failures (A-15).

Limitations. One competition, one model family: no transfer is claimed. Archive A is retrospective, filtered, and its prompt comparison is small and unrandomised. Archive B covers public artifacts only. Our sampling is small — at most one scored result per package, one still pending — so each score is one draw against a best-of-n statistic. The two local blocks differ from the hidden split in task sets, hardware and budget; the binomial calculation is illustrative.

Planned: repeated byte-identical submissions (between-draw variance), difficulty-stratified paired evaluation, and an instrumented runtime separating charged from uncharged calls — planned, not findings.

6 Conclusion

Reading a high score as capability, or a failed run as failure, does not survive our data. The partner archive suggests the leaderboard is not a failure diagnosis either: a position is one output of an unobserved sampling process, informative but not a capability estimate and not a failure diagnosis.

Reproducibility and Data Statement

Authorship and disclosure. This paper merges two independently collected archives from both teams; the framing of §1–§2 follows two draft write-ups that the partner team shared with us on 2026-11-08 (not yet submitted; both are separate research reports prepared within the same team for this track), listed with titles and Table 5 marks their submission status. We did not rerun the partner's experiments; Part A statements are cited by line number from the co-author manuscript, and their numbers are quoted as reported rather than recomputed. This manuscript has been submitted once, to this track, and is not under review elsewhere. The competition artifacts we submitted are the three listed in the submission-provenance table (Table 2). Analysis and drafting used AI agents.

Data and environment. Archive A numbers are in the co-author manuscript transferred 2026-11-08 (SHA-256 5a56b5bc…); raw traces are not redistributed. Archive B claims map to `evidence.csv` rows (source, command, n, value); the 2026-11-03 snapshot is third-party (SHA-256 629e5059abe5…); figures 1–2 are redrawn from these, figure 3 from local blocks; `recompute.py` and summaries will be released. Environment: official `swegemma` harness, Kaggle 4×L4, official CLI; the `58056163` upload digest is unrecorded (PROV-01b), marked not measured.

 Evidence package (claim table, recompute script, figures): https://www.kaggle.com/datasets/tianchan11/gemma4-paper-evidence. Part B claims: `evidence.csv`,

B-subset SHA-256 `88ef3e00…` (PART_B_FROZEN). Submission statement: above (one submission; Table 2).

Table 5. Partner draft write-ups cited for the framing of §1–§2 (submission status as of 2026-11-08).

Writeup	URL
When "Retry Once" Is Not Enough: Editing Failure Loops in a Gemma Coding Agent	not yet submitted (slug withheld)
When "Do Not Use" Matches Tool Removal: A Gemma Graph-Tool Case Study	not yet submitted (slug withheld)

References

1. Jimenez et al. SWE-bench. ICLR 2024. arXiv:2310.06770.
2. Yang et al. SWE-agent. arXiv:2405.15793.
3. Xia et al. Agentless. arXiv:2407.01489.
4. Miller. Adding Error Bars to Evals. arXiv:2411.00640.
5. Madaan et al. Self-Refine. arXiv:2303.17651.
6. Shinn et al. Reflexion. arXiv:2303.11366.
7. Huang et al. LLMs Cannot Self-Correct Reasoning Yet. ICLR 2024. arXiv:2310.01798.
8. Stechly, Valmeekam & Kambhampati. Self-Verification Limitations. arXiv:2402.08115.
9. Berkeley Function Calling Leaderboard (BFCL). gorilla.cs.berkeley.edu; ICML 2025 poster.
10. Qin et al. ToolLLM. arXiv:2307.16789.
11. Chiang et al. Chatbot Arena. arXiv:2403.04132.
12. Singh et al. The Leaderboard Illusion. arXiv:2504.20879.
13. Competition documentation and `swegemma` harness guide.
14. Kamoi et al. Can LLMs Correct Their Own Mistakes? TACL 2024. arXiv:2406.01297.
15. Agarwal et al. Deep RL at the Edge of the Statistical Precipice. NeurIPS 2021. arXiv:2108.13264.
16. Bouthillier et al. Accounting for Variance in ML Benchmarks. MLSys 2021. arXiv:2103.03098.